

ECO 362: Financial Investments

Introduction

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Two goals

1. Even if this is the only finance course that you take, develop basic knowledge to guide personal financial decisions.
2. If you plan to do a finance minor, develop a solid foundation for more advanced courses.

Will this course help you get rich?

- Not exactly.
- The purpose is to develop knowledge and tools to help you better understand financial markets and investments.
- However, some of this knowledge can prevent you from making common financial mistakes.
 - Diversifying and rebalancing your portfolio to maximize returns and minimize risk.
 - Avoiding mutual funds with high fees.
 - Finding mutual funds that outperform the market.
 - Buying enough life insurance.
 - Increasing overall returns by tax efficient investing.

What to expect

- Finance is a quantitative field. This course is not for students who don't enjoy math and numbers.
 - Concepts from probability, statistics, calculus (differentiation), and linear algebra.
 - Computations in Excel that involve some data.
- Problem sets and exams are based entirely on lecture notes (no textbook).
- Even though the concepts may be challenging, I will try to keep the pace slow enough so that you can follow if you pay attention.
- If you feel that I'm going too fast, please stop me!
- Ask questions! If you don't understand something, it's quite likely that there are other students who don't either.

U.S. income and wealth distribution

Percentile of income	Mean income (thousand \$)	Mean net worth (thousand \$)
Less than 20	23	130
20–40	45	219
40–60	73	385
60–80	122	637
80–90	201	1,265
90–100	692	6,630

Source: 2022 Survey of Consumer Finances

U.S. household assets

Type of asset	Percent participation	Conditional mean (thousand \$)
All assets	100	1,194
Financial assets	99	511
Transaction accounts	99	63
Certificates of deposit	7	99
Savings bonds	6	36
Bonds	1	961
Stocks	21	404
Mutual funds	12	961
Retirement accounts	54	334
Life insurance	16	55
Other managed assets	6	518
Nonfinancial assets	92	742
Vehicles	87	40
Primary residence	66	471
Other residential property	13	519
Equity in nonresidential property	6	468
Business equity	15	1,623

Source: 2022 Survey of Consumer Finances

U.S. household liabilities

Type of liability	Percent participation	Conditional mean (thousand \$)
All liabilities	77	164
Mortgages		
Primary residence	42	212
Other	4	243
Lines of credit	2	127
Installment loans		
Education loans	22	47
Vehicle loans	35	21
Other	19	10
Credit card balances	45	6

Source: 2022 Survey of Consumer Finances

Questions about personal finances

- How much should I save?
- Should I keep my money in bank deposits or invest in stocks and bonds?
- Which stocks should be in my portfolio?
- Should I have a retirement account?
- How much should I borrow to buy a house?
- Should I buy life insurance?

Tools for answering these questions

- Knowledge about financial instruments and institutions.
 - What are stocks, bonds, mortgages, life insurance, and derivatives?
 - What is the difference between ETFs and actively managed mutual funds?
 - What is the difference between traditional and Roth IRA or 401 accounts?
- Knowledge of basic principles of financial economics.
 - What absence of arbitrage implies for pricing of securities.
 - Understanding the tradeoff between risk and return.
 - Rewards for bearing systematic risk through factor exposures.
 - Life-cycle theory of saving.

Normative vs. positive economics

- **Normative economics:** Theories of how households *should* behave (e.g., utility maximizing with rational expectations).
 - For example, household should never hold non-diversified portfolios with only a few stocks.
- **Positive economics:** Models that describe how households actually behave, building on insights from psychology and related fields.
 - Some households have non-diversified portfolios or don't even have any stocks at all.
- This course is about normative economics.
- Normative economics is a good starting point, but always ask whether the assumptions underlying the theory are realistic.